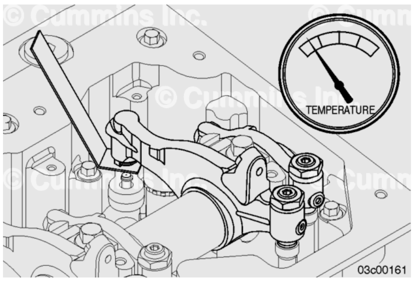
Trainings

General Information

All overhead lash measurements **must** be made when the engine is cold. A stabilized coolant temperature **must** be at 60°C [140°F] or below.

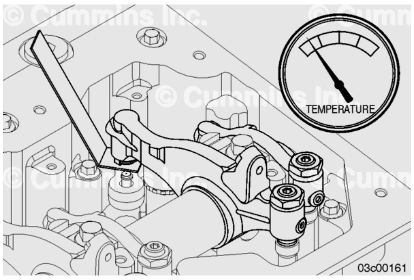


Preparatory Steps

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/269/269-003-011-om.html)

Adjust

Valves and engine brakes **must** be correctly adjusted for the engine to operate efficiently. Valve and engine brake adjustment **must** be performed using the values listed in this section. The accompanying table gives the adjustment specifications.

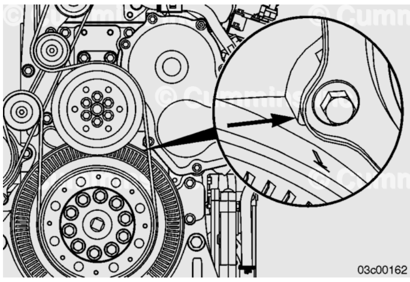


If the valves have been adjusted during troubleshooting or before this scheduled interval, adjustment is **not** required at this time.

Valve Adjustment Specifications			
	mm		in
Intake Valve	0.45	NOM	0.018

Exhaust Valve	0.72	NOM	0.028

All valve and injector adjustments **must** be made when the engine is cold (stabilized coolant temperature at 60°C [140°F] or below).



⚠ WARNING ⚠

When servicing the engine, do not use a remote starter switch or other means that would engage the starting motor to rotate the crankshaft on a engine with a high-pressure common rail fuel system. Using the starting motor can create highly pressurized fuel in the fuel system. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death. Use a hand barring tool to rotate the crankshaft for servicing the engine. Always loosen the pump-to-rail fuel line at the rail to vent the pressure after rotating the crankshaft. Keep hands clear of the line when loosening, and wear appropriate eye protection.

⚠ WARNING ⚠

Do not straighten a bent fan blade or continue to use a damaged fan. A bent or damaged fan blade can fail during operation and cause personal injury or property damage.

The valve set marks are located on the vibration damper. The marks align with a pointer on the gear cover.

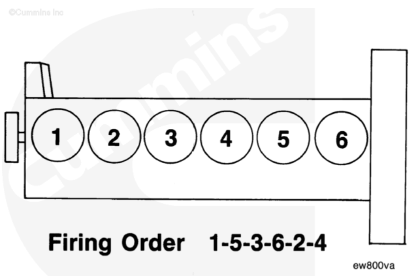
The pointer may be difficult to see. The pointer is located on the bottom left corner of the front gear housing. It may be helpful to use a Dykem™ pen or paint pen to mark the pointer for visibility.

Use the vibration damper to rotate the crankshaft.

The crankshaft rotation is **clockwise** when viewed from the front of the engine.

The cylinders are numbered from the front gear housing end of the engine.

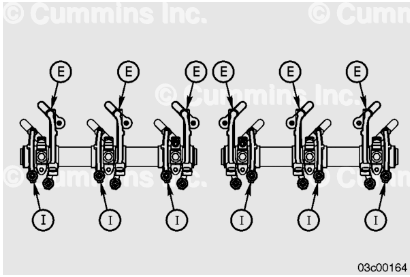
The firing order is 1-5-3-6-2-4.



Each cylinder has two rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The short rocker lever (I) is the intake lever.

Use the accompanying illustration for the intake and exhaust valve rocker lever location identification.



Two crankshaft revolutions are required to adjust all of the valves.

Note : See the example before attempting to begin the adjusting procedure.

Adjust all of the valves to the following valve adjustment sequence chart.

Use a permanent marker to mark each set of valves as they are adjusted.

Valve Measurement Sequence		
Bar engine in direction of rotation	**Damper position**	**Set Cylinder**
Start	A	1
Advance to	B	5
Advance to	C	3
Advance to	A	6
Advance to	B	2
Advance to	C	4
The firing order is 1-5-3-6-2-4		

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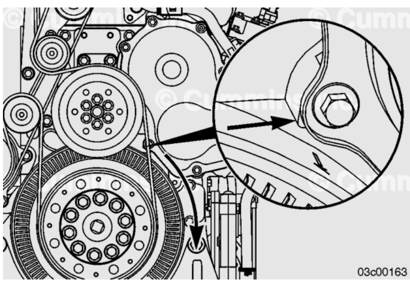


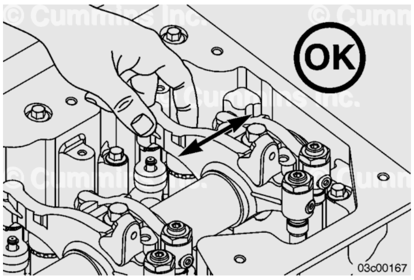
The adjustment can begin on any valve set mark. In the following example, the adjustment will begin on the "A" valve set mark with cylinder number 1 valves closed and ready for adjustment.

Rotate the crankshaft **clockwise** until the "A" valve set mark on the vibration damper is aligned with the pointer on the front gear housing.

The pointer may be difficult to see. The pointer is located on the bottom left corner of the front gear housing. It may be helpful to use a Dykem™ pen or paint pen to mark the pointer for visibility.

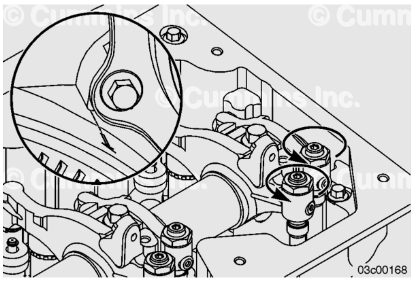
Use the vibration damper to rotate the crankshaft.

When the "A" mark is aligned with the pointer, the intake and exhaust valves for cylinder number 1 **must** be closed. If these conditions are **not** correct, cylinder number 6 valves **must** be ready to set. Set the valves on the cylinder so that both the intake and exhaust valve rocker lever arms are loose and can be moved from side-to-side.

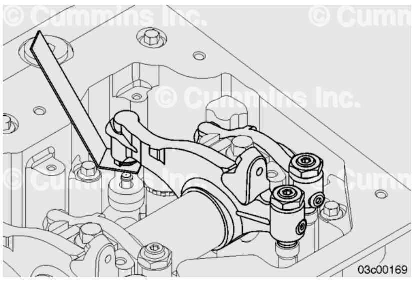


Both valves are closed when both rocker levers are loose and can be moved from side-to-side.

With the "A" valve set mark aligned with the pointer on the front gear housing and both valves closed on the cylinder to be adjusted, loosen the adjusting screw locknuts on the intake and exhaust valves.



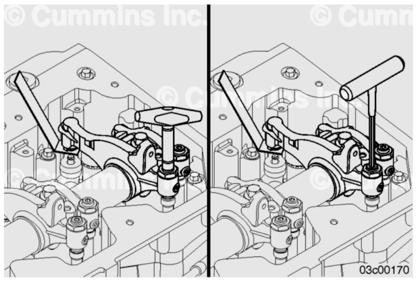
Select a feeler gauge for the correct valve lash specification at ambient temperature.



Valve Lash Specifications			
	mm		in
Intake Valve	0.45	MIN	0.018
Exhaust Valve	0.72	MIN	0.028

Insert the feeler gauge between the top of the crosshead and the rocker lever foot.

Two different methods for establishing valve lash clearance are described below. Either method can be used; however, the torque wrench method has proven to be the most consistent. It eliminates the need to feel the drag on the feeler gauge.



Torque Wrench Method

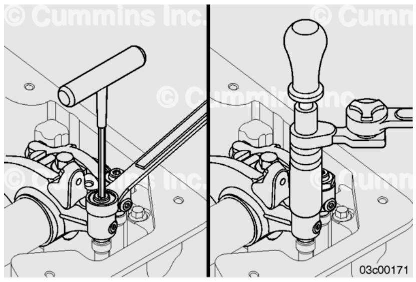
- Use the inch-pound torque wrench, Part Number 3376592, normally used to set preload on top-stop injectors, and tighten the adjusting screw.

Torque Value: 0.7 n•m [6 in-lb]

Touch Method

- Tighten the adjusting screw with an Allen head wrench until a slight drag is felt on the feeler gauge.

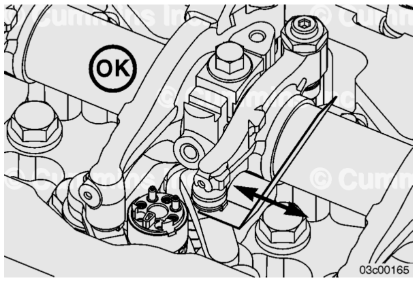
Hold the adjusting screw in the position shown. The adjusting screw **must not** turn when the locknut is tightened.



Torque Value:
Without Torque Wrench Adapter: 61 n•m [45 ft-lb]

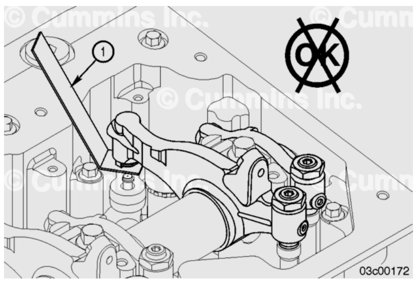
Torque Value:
With Torque Wrench Adapter, Part Number 3163196: 47 n•m [35 ft-lb]

After tightening the locknut to the correct torque value, check to make sure the feeler gauge will slide backward and forward between the crosshead and the rocker lever with **only** a slight drag.



Use a permanent marker to mark each set of valves as they are adjusted.

If using the touch method, attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker between the crosshead and the rocker lever foot. The valve lash is **not** correct when a thicker feeler gauge will fit.



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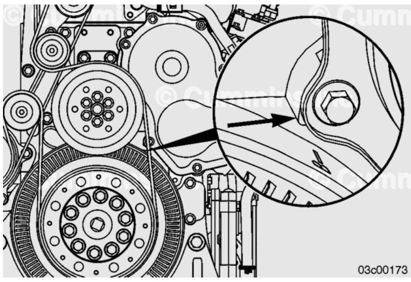

After adjusting the valves and engine brakes, if equipped, on the appropriate cylinder, rotate the crankshaft using the vibration damper, and align the next valve set mark with the pointer on the front gear housing.

The pointer may be difficult to see. The pointer is located on the bottom left corner of the front gear housing. It may be helpful to use a Dykem™ pen or paint pen to mark the pointer for visibility.

Finishing Steps

- Install the rocker lever cover. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/269/269-003-011-om.html)
- Operate the engine. Check for proper operation and leaks.